data into training, testing and validation methods.

4. **Keras:** It is a high performing, user-friendly open sourced python library specifically designed for machine and deep learning tasks.
5. **Pandas:** It is a software library written for Python, and it is used for data manipulation and analysis. It offers data structures and operations for manipulating time series data and statistical tables.
6. **Random:** This python function will create random numbers that will be used when we split or shuffle our data set.
7. **os:** It is an inbuilt python package for accessing your computer and file system. This can be used to display content in directories, create new folders and even delete folders.
8. **gc:** It stands for garbage collector. It is an important tool for manually cleaning and deleting random variables. It will be actively used especially on online kernels such as Kaggle or google colab because they provide limited free memory that may get full since due to the highly computationally expensive task.

Collecting Dataset:

We will be using an existing data set of dogs and cats that are available on the Kaggle platform. The dataset can be found here: *https://www.kaggle.com/c/dogs-vs-cats-redux-kernels-edition/data.* You need to create an account on the Kaggle website to download or use the data. The dataset contains two folders – the train and test folder. The train folder contains 25,000 images of dogs and cats. Each image in the train folder has the label – 1 for dog and 0 for cat as part of the filename. The test folder contains 12,500 images that are named according to a numeric id. Our main task is that for each image in the test set, we should predict a probability that the image is a dog.

Kaggle hosts some of the biggest data science competitions. It is a great place for getting open source data as well as learning from winning experts. For the people from Africa, an alternative platform called Zindi - *http://zindi.africa/* exists that is similar to Kaggle, but it is fine-tuned to the African society. It contains data